Experiment 3: Handle Missing and Categorical Data using Pandas & Scikit-learn

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Aim / Objective

To learn and apply techniques for handling missing values and encoding categorical features using Pandas and Scikit-learn.

Theory / Introduction

In real-world datasets, it is common to encounter missing values and categorical variables. Handling them properly is essential for accurate machine learning models.

- **Missing Values**: Can be handled by removing or imputing (mean, median, mode).
- **Categorical Features**: Should be encoded into numerical format for ML models. Common techniques include:
    - **Label Encoding**
    - **One-Hot Encoding**

In this experiment, we will use the **Titanic dataset** to demonstrate how to preprocess missing and categorical data before applying machine learning algorithms.

Software / Tools Required

- Python
- Pandas
- Scikit-learn

Dataset Used

Titanic Dataset

About the Dataset

The **Titanic dataset** contains data of passengers aboard the Titanic ship. It includes personal and travel information such as age, sex, passenger class, fare, and survival status.

Key features:

- Age and Embarked columns contain missing values
- Sex and Embarked are categorical variables

🔗 Kaggle Titanic Dataset

Procedure / Steps

1. Load the dataset
2. Identify missing values using .isnull()
3. Handle missing values using imputation or deletion
4. Encode categorical columns (Sex, Embarked) using One-Hot Encoding
5. Return a clean DataFrame ready for ML training

Python Step-by-Step Code with Explanations

python

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```
# Step 1: Import libraries
import pandas as pd
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder

# Step 2: Load Titanic dataset
df = pd.read_csv("https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv")
print("Initial shape:", df.shape)

# Step 3: Select columns of interest
data = df[['Age', 'Sex', 'Embarked', 'Fare']]
print("\nMissing values before handling:")
print(data.isnull().sum())

# Step 4: Impute missing 'Age' with mean and 'Embarked' with mode
imputer = SimpleImputer(strategy='mean')
data.loc[:,'Age'] = imputer.fit_transform(data[['Age']])[:,0]

imputer_mode = SimpleImputer(strategy='most_frequent')
data.loc[:, 'Embarked'] = imputer_mode.fit_transform(data[['Embarked']])[:, 0]
```

```
print("\nMissing values after imputation:")
print(data.isnull().sum())

# Step 5: Encode categorical columns
data_encoded = pd.get_dummies(data, columns=['Sex', 'Embarked'])
print("\nFinal DataFrame after encoding:")
print(data_encoded.head())
```

Result / Output

- Missing values in 'Age' and 'Embarked' columns were filled
- Categorical variables (Sex, Embarked) were successfully encoded into numeric form using **One-Hot Encoding**
- Final DataFrame is ready for ML model training

Conclusion

The experiment demonstrated how to identify and handle missing values and encode categorical variables using Pandas and Scikit-learn. Proper preprocessing improves the performance of machine learning models and avoids common training issues.

Viva Questions (2–3 per experiment)

1. What are different strategies for imputing missing values?
2. When would you use One-Hot Encoding over Label Encoding?
3. What are the consequences of ignoring missing data?